



# Experimental Investigation of Self-Healing Concrete Incorporating Bacterial and Nano-Silica Additives

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**Abstract:** The problem of cracking gives rise to one of the most serious issues associated with the durability of reinforced concrete. To overcome this shortcoming, the objective of this study involved an experimental investigation of the mechanical properties and self-healing capability of concrete in the presence of bacteria and nanoparticles of silica. Five different mixes had been considered in the experiment and they were: (i) reference mix; (ii) bacterial concrete; (iii) nano-silica concrete; (iv) concrete hybrids made of bacteria and (v) of nano-silica in various ratios. Surprisingly, the mechanical properties and the durability of concrete were largely enhanced, given the existence of bacteria and nanoparticles of silica. A mix of concrete with 4% nanosilica and 106 cells/ml of bacteria indicated maximum compressive strength, which equaled to 58.2 MPa within 56 days compared with 40.2 MPa of the reference sample (44.8% improvement). Meanwhile, the splitting tensile strength could be improved up to 38.2%. Bacteria and silica improve pores refining, resulting in increased matrix density. As regards self-healing ability, impressive results were found in concrete hybrids. Crack-closure of the reference sample was 26.2%, while that of hybrid mix was 93.3%. Scanning electron microscopy (SEM) analysis revealed that hybrid samples included calcium carbonates and C-S-H gel in cracks, whereas self-healing contributed to 95.6% concrete strength restoration. Interestingly, the combined effects of bacteria and nano-silica could help manufacture highly durable self-healing concrete for the construction industry.

**Keywords:** Self-healing concrete; Bacterial concrete; Nano-silica; Crack healing; Calcium carbonate precipitation; Durability

## 1 Introduction

Concrete is one of the most widespread construction materials worldwide due to its abundance, low price, and high compressive strength [1]. However, its brittleness renders concrete prone to cracking caused by shrinkage, thermal stresses, loading, and other external influences [1]. Cracking is detrimental for concrete durability and lifespan; it leads to water, chlorides, sulfates, and other aggressive agents penetrating concrete, resulting in more bar corrosion and further damage [2]. Due to its energy-intensive nature, cement manufacturing markedly contributes to the emission of CO<sub>2</sub> into the environment. Therefore, the necessity of enhancing the longevity of buildings by using self-healing techniques is a vital requirement for reducing the impacts of concrete rebuilding due to repair [3].

Recently, the self-healing concrete technology has been considered a promising way to tackle concrete durability-related issues [4]. Self-healing concrete is capable of sealing micro-cracks autonomously, hence increasing its durability and extending its lifespan [5]. Self-healing mechanisms include autogenous healing, encapsulated polymers, mineral admixtures, bacterial concrete, etc. [6]. Among these methods, bacterial self-healing concrete is one of the most environmentally friendly and efficient inventions in terms of crack sealing [7].

The basis of bacterial concrete lies in microbial-induced calcium carbonate precipitation (MICP). According to this method, certain bacteria precipitate calcite crystals in cracks and pores [8]. The bacteria used in microbial-induced calcite precipitation include the genus *Bacillus* such as *Bacillus subtilis* (*B. subtilis*), *Bacillus sphaericus* (*B. sphaericus*), and *Sporosarcina pasteurii* (*S. pasteurii*), since these have been proved to survive the harsh conditions in concrete due to their tolerance of high pH levels, and they could produce urease required for the formation of calcite [9]. CaCO<sub>3</sub> formation is useful in sealing cracks and improving the mechanical strength of concrete [10]. Previous literature reported that MICP could seal cracks 0.8 mm in width and demonstrate increased durability [11].

Apart from bacterial additives, another area of research that has brought remarkable achievements in recent years is nanotechnology. Nano-silica, nano-alumina, carbon nanotubes, etc., were used as nano-scale additives to optimize cementitious composite performance [12]. Nano-silica stands out due to its high pozzolanic activity and fine particle size. Thus, the performance of concrete microstructure and mechanical characteristics could be largely enhanced [13]. As a consequence of interacting with calcium hydroxide released during cement hydration, the particles of nano-silica produce additional C-S-H gel, leading to densification of the matrix and refinement of the pore structure [14]. Nano-silica is known to increase the compressive strength, tensile strength, durability, and chloride resistance of concrete [15].

Numerous studies have been devoted to developing self-healing concrete based on bacterial agents and nanoparticles [16]. The incorporation of biomineralization and densification mechanisms could help achieve a combined effect and improve the crack-healing efficiency of the composite and its mechanical characteristics [17]. While nano-silica provides nuclei for the hydration reaction and bacteria facilitate the formation of calcite, they together decrease water absorbance and improve the microstructure and mechanical performance of concrete [18, 19].

In addition, the utilization of nanoparticles improves bacterial survival in concrete due to pore refinement and improvement of microenvironmental conditions [20]. Experiments revealed that hybrid concrete based on bacteria and nano-silica had higher compressive strength and crack-healing efficiency than concrete with only bacteria included [21]. Scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS) analysis have revealed the presence of dense deposition of calcium carbonate and stronger bonding within the interfacial layer [22]. Hybrid self-healing concrete exhibits improved performance against freeze-thaw cycling, sulfate corrosion, and chloride intrusion [23].

Despite significant advancement within the area of self-healing concrete technologies, there still exist a number of issues, such as improvement of bacterial and silica nanoparticle concentration as well as crack healing efficiency [24]. While many previous studies were related to either bacterial concrete or nano-modified concrete alone, few of them examined the influence of bacteria and nanoparticles on mechanical and self-healing properties of concrete simultaneously [25]. Therefore, further experiments should be designed to explore the combined effect of bacteria and nanoparticles on the concrete properties as a research field.

In this context, previous studies primarily focused on the individual impacts of nano-silica and bacterial concentration on the properties of the samples under certain parameters. Existing studies discovered that using nano-silica at levels of 2% and 4% was optimal for increasing the pozzolanic activity of the mixture, thus creating denser microstructure while avoiding agglomeration. On the other hand, bacterial concentration of  $10^5$  to  $10^6$  cells/mL proved to be an effective threshold range for triggering the MICP process within the cracks, without deteriorating the density of the material. Nonetheless, there is an unexplored area of research regarding the interaction between these values of parameters, namely, a combination of the upper limit values of nano-silica and bacteria concentration, which is a 4% content of the nano-silica in a concentration of  $10^6$  cells/mL. There is a lack of research documenting the concomitant effects from the interaction between the advanced sealing process and microenvironmental conditions of nano-silica on the viability of bacteria concentration. Therefore, such a choice of parameters in the current paper was not accidental.

Simply put, the objective of this study is to experimentally investigate the influence of bacterial additives and nano-silica on the fresh properties, mechanical performance, durability, and self-healing capability of concrete. Mixtures of concrete with bacterial additives, nano-silica, and their hybrids were studied with respect to compressive strength, splitting tensile strength, water absorbance, rapid chloride permeability test, and self-healing efficiency. In addition, microstructural analyses using SEM and EDS were carried out to focus on the calcium carbonate precipitation and densification processes.

## **2 Methodology**

### **2.1 Materials**

#### **2.1.1 Cement**

Ordinary Portland Cement Type I, which meets American Society for Testing and Materials (ASTM) C150 standards, was used in this research. The density and Blaine fineness of cement were 3.15 and about 340 m<sup>2</sup>/kg, respectively. Chemical compositions of cement mainly consisted of calcium oxide (CaO), silica (SiO<sub>2</sub>), alumina (Al<sub>2</sub>O<sub>3</sub>), and iron oxide (Fe<sub>2</sub>O<sub>3</sub>).

#### **2.1.2 Fine aggregate**

River sand as fine aggregate in accordance with ASTM C33 was selected for this research. The fineness modulus and specific gravity of fine aggregate were 2.71 and 2.63, respectively, and the maximum particle size was 4.75 mm.

#### **2.1.3 Coarse aggregate**

Gravel with nominal maximum size of 12.5 mm was adopted as coarse aggregate for concrete mixture. Coarse aggregate was selected based on ASTM C33 standard with specific gravity of 2.68 and water absorption of 0.9%.

#### 2.1.4 Water

Tap water without any impurities, oil, and organic materials was used as mixing and curing water of concrete mixture.

### 2.2 Bacterial Culture

The highly resistant bacteria, *B. subtilis*, was selected due to its ability to survive the harsh alkaline condition and produce calcium carbonate crystals. Specifically, *B. subtilis* was utilized due to its capacity to form robust endospores. These spores remain in a dormant state, allowing them to successfully withstand the severe mechanical shear during concrete mixing and the highly alkaline internal environment of the cementitious matrix. They are reactivated to induce mineralization only when moisture and oxygen ingress through subsequent crack propagation. *B. subtilis* was cultured in a broth consisting of peptone, yeast extract, sodium chloride and urea, and then incubated in an incubator at 30 °C for 24 hours until concentration of  $10^5$ – $10^6$  cells/mL was reached. Bacillus suspension was added into the mixing water for concrete mixture. Ca-lactate was introduced into concrete as a precursor of forming calcium carbonate crystals.

Calcium lactate was specifically selected as the source of organic carbon because, unlike alternative nutrients or sugars, it does not impede or retard the early hydration reactions of cement, thereby ensuring that the development of initial mechanical strength would not be compromised.

### 2.3 Nano-Silica

Nano-silica powder, with an average size of 15–20 nm and more than 99% purity, was utilized for this experiment. The specific surface area of nano-silica was found to be around 200 m<sup>2</sup>/g. Nano-silica was utilized to partially replace the cement content by mass at levels of 2% and 4%. Before mixing, nano-silica particles were mixed with water for 15 minutes by means of an ultrasonic mixer in order to avoid agglomeration of the particles.

### 2.4 Design of Concrete Mixes

Experimental results are of great interest both for application and scientific research. Five different concrete mixes were applied in this experiment, including one control mix and four modified mixes with bacterial addition and nano-silica. The water-cement ratio (w/c) remained unchanged at 0.45 for all concrete mixes. The composition of the concrete mixes is listed in Table 1.

**Table 1.** Mix proportions of concrete mixtures

| Mix ID           | Nano-Silica (%) | Bacterial Concentration (cells/mL) | Water-Cement Ratio (w/c) |
|------------------|-----------------|------------------------------------|--------------------------|
| C0 (Control)     | 0               | 0                                  | 0.45                     |
| B1 (Bacteria)    | 0               | $10^5$                             | 0.45                     |
| N1 (Nano-silica) | 2               | 0                                  | 0.45                     |
| BN1 (Hybrid 1)   | 2               | $10^5$                             | 0.45                     |
| BN2 (Hybrid 2)   | 4               | $10^6$                             | 0.45                     |

For research purposes, the BN2 mixture was designed to combine the highest dosage of nano-silica (4%) with the maximum bacterial concentration ( $10^6$  cells/mL). The rationale behind this specific grouping, apparently, was to evaluate the upper-bound combined potential of the hybrid system under maximum modification levels, rather than isolating or decoupling the individual effects of each additive.

### 2.5 Specimen Preparation

Concrete was mixed using a laboratory pan mixer with a capacity of 0.05 m<sup>3</sup>. Cement, nano-silica, fine and coarse aggregates were first dry-mixed for about 3 minutes. Water and bacterial suspension were gradually added while mixing for another 5 minutes. Concrete specimens were cast in three layers in steel molds using a vibrating table to ensure adequate compaction. They were allowed to rest under a plastic cover for 24 hours before removing from the molds. After removal, all specimens were placed in water curing at  $25 \pm 2$  °C for specified testing periods.

### 2.6 Experimental Program

The experimental program involved evaluating the properties of the concrete including fresh concrete, durability, mechanical and self-healing properties.

#### 2.6.1 Slump test

The ability of the fresh concrete mix to flow under its own weight was determined through slump cone test. It was carried out according to standard procedure ASTM C143.

### 2.6.2 Compressive strength test

The compressive strength test was done on cubic samples with dimensions of 150 × 150 × 150 mm in accordance with ASTM C39. Specimens were tested at different curing ages, i.e., at 7, 28, and 56 days with a hydraulic compression tester having a loading speed of 0.25 MPa/s. Compressive strength was measured by the following formula:

$$f_c = \frac{P}{A} \quad (1)$$

where,

$f_c$  = compressive strength (MPa)  
 $P$  = maximum applied load (N)  
 $A$  = loaded surface area (mm<sup>2</sup>)

### 2.6.3 Splitting tensile strength test

The splitting tensile strength test was carried out on cylindrical specimens of 150 × 300 mm in accordance with ASTM C496 up to 28 days. Splitting tensile strength was measured by:

$$f_t = \frac{2P}{\pi LD} \quad (2)$$

where,

$f_t$  = splitting tensile strength (MPa)  
 $P$  = applied load (N)  
 $L$  = specimen length (mm)  
 $D$  = specimen diameter (mm)

### 2.6.4 Water absorption test

Tests for water absorption followed ASTM C642 standard and were carried out to analyze the porosity and permeability characteristics of the concrete. Wetting and drying methods were used to determine water absorption percentage as follows:

$$WA = \frac{W_s - W_d}{W_d} \times 100 \quad (3)$$

where,

$WA$  = splitting tensile strength (MPa)  
 $W_s$  = applied load (N)  
 $W_d$  = specimen length (mm)

### 2.6.5 Rapid chloride permeability test

Rapid chloride permeability tests were performed following ASTM C1202 to assess the capacity of concrete mixtures to withstand the penetration of chloride ions. The total electrical charge passed through the sample during a 6-hour testing period was calculated in coulombs.

### 2.6.6 Self-healing evaluation

To evaluate the self-healing ability, artificially-induced cracks in concrete specimens were assessed in terms of their resistance to further deterioration in a 28-day wet/dry cycle. Initially, cracks with the width of 0.3 to 0.5 mm were generated using a flexural loading system after 28 days of curing. Afterwards, specimens were cured by wet/dry method for 28 days. Before and after the healing process, cracks widths were analyzed using microscopic images. The healing ratio was then calculated as follows:

$$HR = \frac{W_i - W_h}{W_i} \times 100 \quad (4)$$

where,

$HR$  = healing ratio (%)  
 $W_i$  = initial crack width (mm)  
 $W_h$  = healed crack width (mm)

### 2.6.7 Microstructural analysis

SEM and EDS techniques were applied to analyze the healing products. Gold-coated samples were obtained from the areas of healed cracks before conducting SEM studies. In addition, EDS analyses were performed to determine the chemical composition.

## 2.7 Statistical Analysis

The tests for each concrete mixture were conducted on at least three specimens. For statistical analysis, one-way analysis of variance (ANOVA) was used to test the effects of bacterial and nano-silica incorporation on concrete properties at 95% confidence level.

## 3 Results

### 3.1 Fresh Concrete Properties

Fresh concrete characteristics of all the tested mixes are provided in Table 2. An evident slump was recorded for all samples with an increase in the nano-silica content and bacterial inclusion. The maximum slump value, i.e., 82 mm, corresponded to control sample C0, while the minimum slump value was registered for hybrid BN2 mix (slump equal to 62 mm).

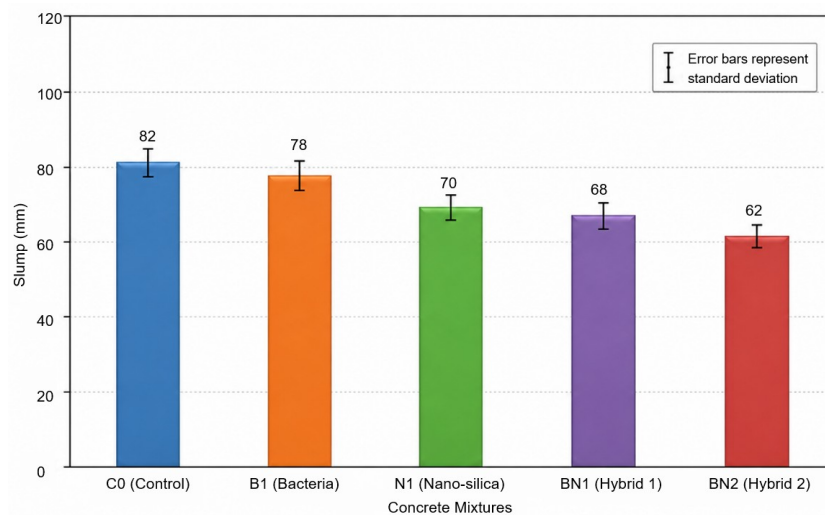
**Table 2.** Fresh concrete properties

| Mix ID           | Slump (mm) | Fresh Density (kg/m <sup>3</sup> ) | Setting Time (min) |
|------------------|------------|------------------------------------|--------------------|
| C0 (Control)     | 82         | 2385                               | 145                |
| B1 (Bacteria)    | 78         | 2390                               | 148                |
| N1 (Nano-silica) | 70         | 2402                               | 152                |
| BN1 (Hybrid 1)   | 68         | 2410                               | 155                |
| BN2 (Hybrid 2)   | 62         | 2420                               | 160                |

It is worth noting that despite the observed decrease, all slump values (ranging from 62 mm to 82 mm) remained aptly within the “medium workability” category based on standard concrete practice. This indicated that the reduction in workability did not compromise practical on-site execution. The concrete mixtures were completely viable for standard handling, casting, and mechanical compaction via vibration in structural applications, to guarantee improved hardened properties without sacrificing the quality of field placement.

An obvious slump reduction was mainly caused by the high specific surface area of nano-silica particles, contributing to higher water consumption and accelerated processes of cement hydration [12–15]. Indeed, studies by Jo et al. [14] and Li et al. [15] proved that adding nano-silica to concrete resulted in significantly reduced workability, as this material has a high water adsorption capability. Besides, the presence of bacterial additives led to a slight increase in the viscosity of fresh concrete due to altered rheological characteristics [8].

Figure 1 demonstrates the variation of slump values for all concrete mixes. Moreover, a small density increase for all samples was recorded and could be explained by the superior concrete matrix compaction caused by nano-silica [13].



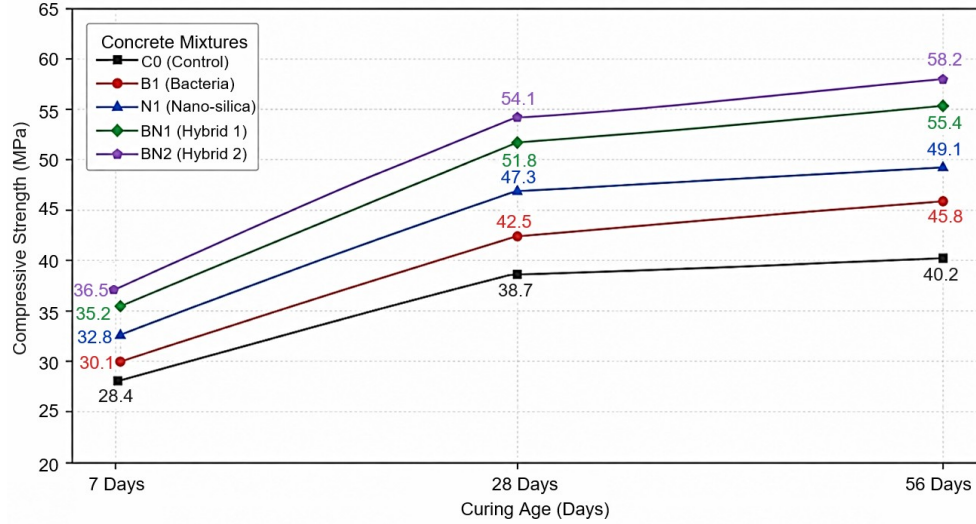
**Figure 1.** Effects of bacterial additives and nano-silica on the slump values of concrete mixtures

### 3.2 Mechanical Properties

The compressive strength results at curing ages of 7, 28, and 56 days are presented in both Table 3 and Figure 2.

**Table 3.** Compressive strength results

| Mix ID           | 7 Days (MPa) | 28 Days (MPa) | 56 Days (MPa) |
|------------------|--------------|---------------|---------------|
| C0 (Control)     | 28.4         | 38.7          | 40.2          |
| B1 (Bacteria)    | 30.1         | 42.5          | 45.8          |
| N1 (Nano-silica) | 32.8         | 47.3          | 49.1          |
| BN1 (Hybrid 1)   | 35.2         | 51.8          | 55.4          |
| BN2 (Hybrid 2)   | 36.5         | 54.1          | 58.2          |

**Figure 2.** Development of compressive strength of concrete mixtures at different curing ages (by Day)

The compressive strengths of all modified concrete mixtures proved higher compared with the control concrete mixture. The hybrid concrete mixture BN2 demonstrated the highest compressive strength of 58.2 MPa after 56 days, representing an increase of about 44.8% from the control mixture.

The increase in the compressive strength could be attributed to the formation of additional C-S-H gel from the reaction between nano-silica and the calcium hydroxide produced during the hydration of cement, leading to better densification of the matrix and lowered pore connectivity [14, 15]. Besides, the bacteria produced carbonates, which precipitate inside the matrix pores and micro-cracks, thereby improving the densification of the matrix and increasing the load transfer efficiency [9, 10].

These results agree with the conclusions of the study carried out by Wang et al. [10], which remarked that significant increases in compressive strength were attributable to calcite precipitation. Furthermore, the findings of Achal et al. [8] indicated that microbial mineralization was capable of considerably increasing compressive strength due to crack filling and refining of pores. Recent studies conducted by Damodarana et al. [18] also confirmed that hybrid mixtures of nano-silica and bacteria contained higher mechanical properties, compared with conventional concrete mixtures.

The test results for the splitting tensile strength after 28 days are provided in the following Table 4.

**Table 4.** Splitting tensile strength results

| Mix ID           | Tensile Strength (MPa) | Improvement (%) |
|------------------|------------------------|-----------------|
| C0 (Control)     | 3.4                    | —               |
| B1 (Bacteria)    | 3.8                    | 11.8            |
| N1 (Nano-silica) | 4.1                    | 20.6            |
| BN1 (Hybrid 1)   | 4.5                    | 32.4            |
| BN2 (Hybrid 2)   | 4.7                    | 38.2            |

The splitting tensile strength dramatically increased as a result of introducing bacteria and nano-silica into the concrete mixture. In terms of tensile strength, the hybrid mixture BN2 obtained the best results, reaching 4.7 MPa. The increase could be explained by the refinement of the interfacial transition zone (ITZ) and reduced microvoids inside the matrix because of adding nano-silica [12, 15]. Besides, bacteria helped bridge the micro-cracks with calcite

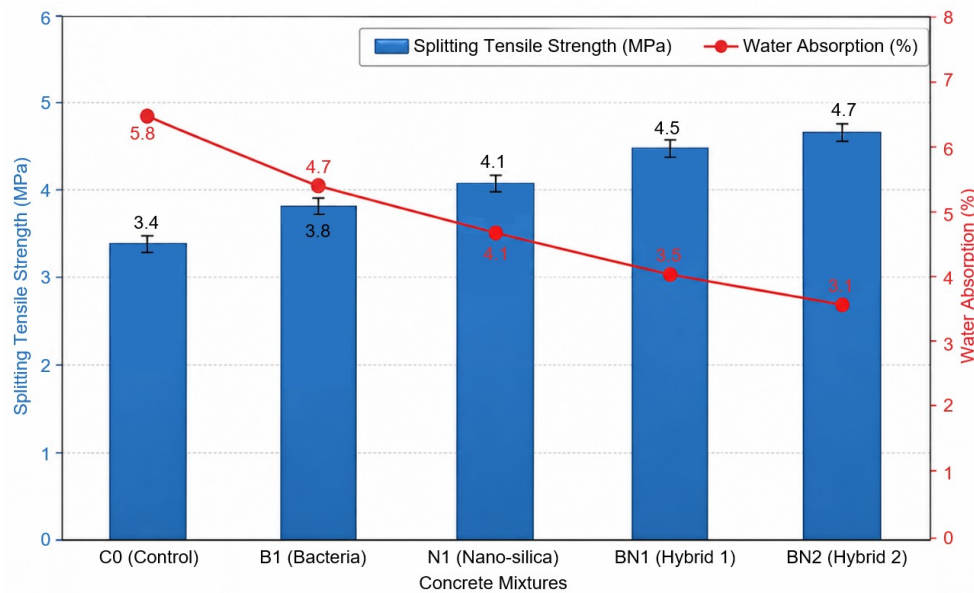
precipitates, which delayed crack propagation under tensile loads [9, 10]. Jonkers et al. [9] reported similar results concerning increased crack resistance and tensile properties in bacterial concrete systems. To concur, Zhang and Ding [22] confirmed that self-healing systems enhanced tensile properties owing to mineral deposition in damaged regions.

### 3.3 Mechanical Properties

The water absorption and rapid chloride permeability test (RCPT) results are summarized in Table 5 and illustrated in Figure 3.

**Table 5.** Water absorption and chloride permeability results

| Mix ID           | Water Absorption (%) | Rapid Chloride Permeability Test (RCPT) Charge Passed (Coulombs) | Permeability Reduction (%) |
|------------------|----------------------|------------------------------------------------------------------|----------------------------|
| C0 (Control)     | 5.8                  | 3850                                                             | —                          |
| B1 (Bacteria)    | 4.7                  | 3100                                                             | 19.5                       |
| N1 (Nano-silica) | 4.1                  | 2550                                                             | 33.8                       |
| BN1 (Hybrid 1)   | 3.5                  | 1900                                                             | 50.6                       |
| BN2 (Hybrid 2)   | 3.1                  | 1450                                                             | 62.3                       |



**Figure 3.** Comparison of splitting tensile strength and water absorption of concrete mixtures

The improved concrete mixes uncovered considerable reduction in water absorption rate and chloride penetration compared with the control. Amongst the above mix samples, mix BN2 recorded the least water absorption percentage of 3.1% and lowest chloride penetration at 1450 Coulombs. From an engineer's perspective, the major decrease in water absorption and total charge passed (for RCPT) has important implications in terms of durability under actual usage. The reduction in permeability means that there will be reduced entry of water, chlorides, sulfates, and other harmful materials into the concrete matrix, resulting in improved corrosion resistance of the reinforcement bars embedded within the concrete structure. Therefore, from the engineering viewpoint, this development has helped improve durability and extend the lifespan of such structures.

This phenomenon could be explained by the improved permeability properties in the formation of refined pores and dense matrices out of the hydration reactions between the formation of nano-silica and bacterial calcite [12, 15, 20]. The presence of calcium carbonate particles in the concrete matrix filled in the pores and limited voids [10]. This finding is consistent with the works by Sanchez and Sobolev [12], who stated that the incorporation of nanomaterials could increase impermeability by densifying microstructure. Likewise, Wang et al. [10] found that bacterial self-healing systems limited chloride ingress through crack filling and capillary porosity.

### 3.4 Self-Healing Performance

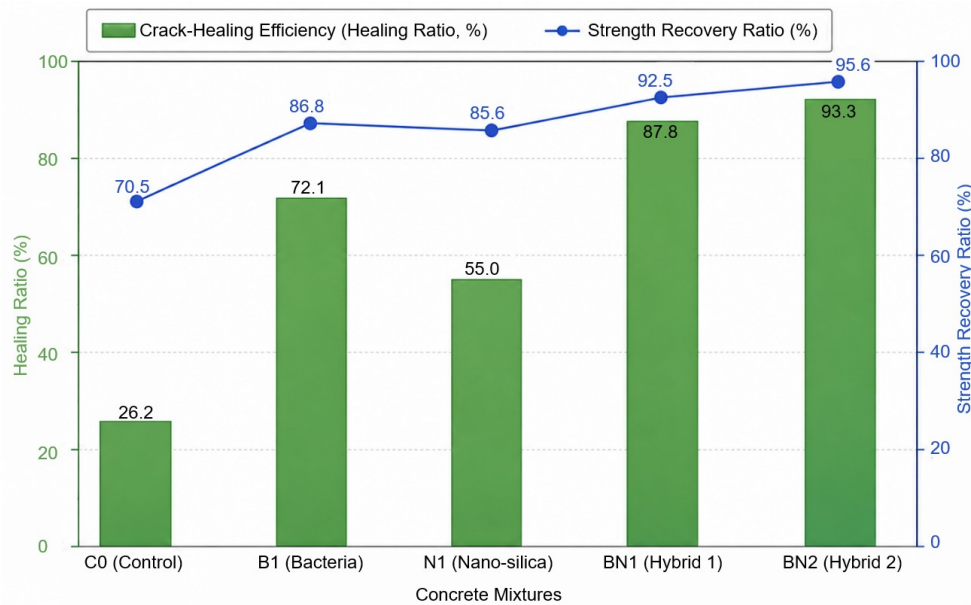
The results on crack healing and strength recovery performance are recorded in Tables 6 and 7, respectively as well as illustrated in Figure 4.

**Table 6.** Crack-healing efficiency

| Mix ID           | Initial Crack Width (mm) | Healed Crack Width (mm) | Healing Ratio (%) |
|------------------|--------------------------|-------------------------|-------------------|
| C0 (Control)     | 0.42                     | 0.31                    | 26.2              |
| B1 (Bacteria)    | 0.43                     | 0.12                    | 72.1              |
| N1 (Nano-silica) | 0.40                     | 0.18                    | 55.0              |
| BN1 (Hybrid 1)   | 0.41                     | 0.05                    | 87.8              |
| BN2 (Hybrid 2)   | 0.45                     | 0.03                    | 93.3              |

**Table 7.** Strength recovery after healing

| Mix ID           | Initial Strength (MPa) | Cracked Strength (MPa) | Recovered Strength (MPa) | Recovery Ratio (%) |
|------------------|------------------------|------------------------|--------------------------|--------------------|
| C0 (Control)     | 38.7                   | 24.1                   | 27.3                     | 70.5               |
| B1 (Bacteria)    | 42.5                   | 27.8                   | 36.9                     | 86.8               |
| N1 (Nano-silica) | 47.3                   | 31.2                   | 40.5                     | 85.6               |
| BN1 (Hybrid 1)   | 51.8                   | 34.6                   | 47.9                     | 92.5               |
| BN2 (Hybrid 2)   | 54.1                   | 36.5                   | 51.7                     | 95.6               |



**Figure 4.** Crack-healing efficiency and strength recovery ratio of concrete mixtures

On the other hand, control concrete indicated only minor signs of self-healing due to autogenous mechanisms related to delayed hydration of cement [5]. Both bioconcrete and hybrid specimens possessed remarkable abilities to close cracks. Out of all concrete mixtures, hybrid concrete mixture BN2 provided the best self-healing efficiency with a healing ratio of 93.3% and regained almost 95.6% of initial compressive strength.

When evaluating self-healing performance, a fundamental distinction has to be maintained between visual crack closure (Table 6) and mechanical strength recovery (Table 7), as they represent completely different physical phenomena. Superficial formation of calcium carbonate may lead to the visual closure of cracks via the blocking of the crack entrance. Nevertheless, it is not an automatic guarantee that the material will regain its mechanical strength. While the high crack healing percentages in the bacterial consortiums and hybrid systems are largely due to the formation of a thin layer of calcite crystals on the crack surfaces, the exceptional strength recovery ability (95.6%) of hybrid system BN2 depends upon a different mechanism. Within the deeper regions of the crack, the

high pozzolanic activity of the 4% nano-silica promotes the continuous generation of additional C-S-H gel. This gel formation works in tandem with the microbial calcite precipitation, to establish robust internal chemical and mechanical bonding across the internal crack walls. Therefore, the combined system could ensure true structural strength restoration deep within the matrix rather than a merely superficial or cosmetic visual seal.

The outstanding behavior was a result of bacteria-induced precipitation of calcium carbonates along crack surfaces in combination with nano-silica pore sealing ability [9, 10, 18]. It has been proven earlier by Zheng et al. [11] that bacterial spores successfully sealed cracks due to calcite precipitation. Apart from this evidence, Delesky et al. [23] stated that biomineralized self-healing techniques notably improved crack closure properties and durability performances. Visual observations revealed that dense layers of white calcite precipitates formed on crack surfaces, indicating the activity of bacterial mineralization.

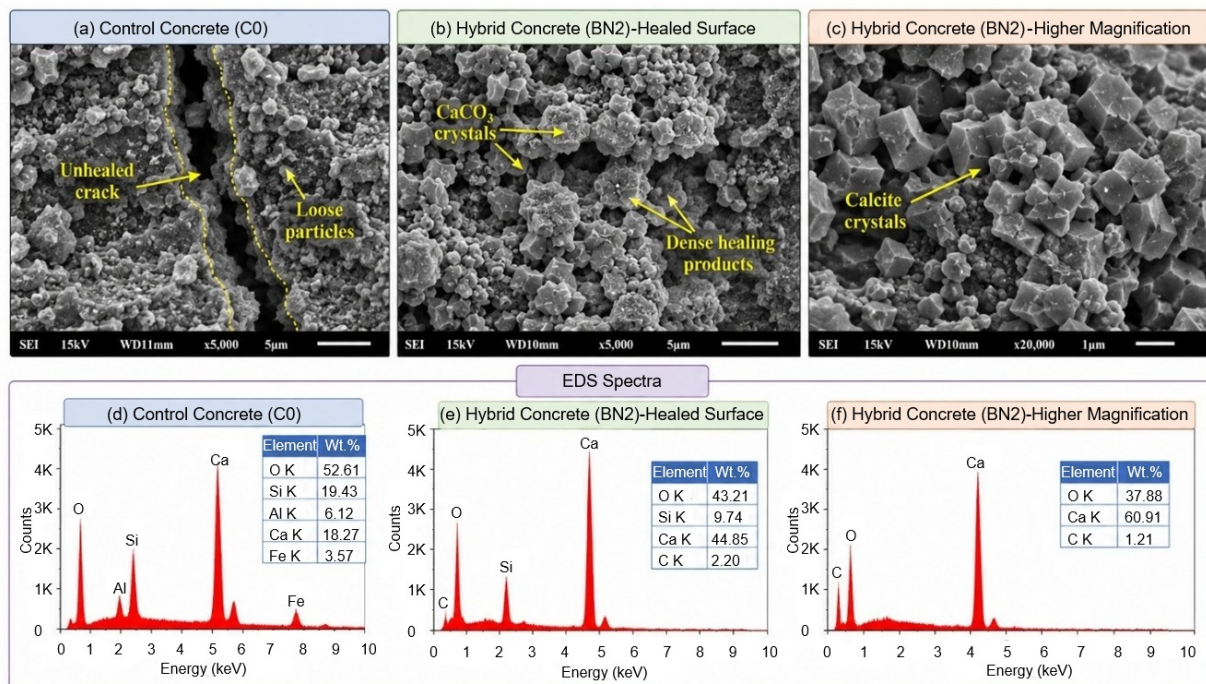
### 3.5 Microstructural Analysis

To examine the structure of healing agents and modification in microstructure, SEM/EDS analysis was performed on concrete samples. Typical results of SEM/EDS examinations are presented in Table 8.

**Table 8.** Microstructural analysis results

| Mix ID           | CaCO <sub>3</sub> Deposition | Pore Density | C-S-H Formation | Crack Closure   |
|------------------|------------------------------|--------------|-----------------|-----------------|
| C0 (Control)     | Low                          | High         | Moderate        | Partial         |
| B1 (Bacteria)    | High                         | Moderate     | Moderate        | Significant     |
| N1 (Nano-silica) | Moderate                     | Low          | High            | Moderate        |
| BN1 (Hybrid 1)   | Very high                    | Very low     | Very high       | Excellent       |
| BN2 (Hybrid 2)   | Excellent                    | Minimal      | Excellent       | Nearly complete |

SEM revealed calcite crystal formations in the repaired fissures of both bacterial and hybrid concrete compositions. In addition, there were distinct peaks of calcium, carbon, and oxygen from EDS spectra, which are consistent with the deposition of calcium carbonate. The hybrid concrete exhibited the highest density of the microstructure due to the combination of mineralizing effect of bacteria and nano-silica hydration activity [18, 22]. The formation of additional C-S-H gel and calcite crystals effectively reduced porosity and improved the compactness of the concrete matrix (Figure 5).



**Figure 5.** Scanning electron microscopy (SEM) micrographs and energy-dispersive spectroscopy (EDS) spectra of healed crack surfaces in hybrid concrete

Note: The presence of strong Ca, C and O peaks in hybrid concrete confirms the formation of CaCO<sub>3</sub> (calcite) crystals that effectively fill and heal cracks

Similar to the work done by Tan et al. [16], the findings of Wong et al. [25] confirmed that the microstructure of hybrid self-healing concrete is considerably more compact and durable, thanks to the combined healing processes.

#### 4 Conclusions

This research experimentally examined the effect of bacteria and nano-silica on various characteristics of concrete including mechanical properties, durability, and self-healing ability. As evidenced from this work, the performance level was substantially higher when bacterial and nano-silica elements were added to concrete mixtures.

The hybrid concrete mixture with a composition of 4% nano-silica and bacteria at the concentration of  $10^6$  cells/mL revealed the highest compressive strength – 58.2 MPa. That was an increase of about 44.8% compared with the control sample. Splitting tensile strength massively increased owing to matrix densification and crack bridging effects arising from the creation of extra calcium carbonate precipitates and C-S-H gel.

To conclude, improved values were found in characteristics like water absorption rate and chloride penetration rate. Hybrid samples were interpreted and found with a higher resistance to aggressive environments due to their compactness. As regards self-healing properties, all tested bacteria and hybrid concrete acquired excellent healing ability. The hybrid concrete mixture excelled with even better self-healing efficiency and recovered compressive strength by 95.6%. Both SEM and EDS examinations revealed the formation of dense calcium carbonate precipitates in healed cracks and pores.

#### Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

#### Conflicts of Interest

The authors declare that they have no conflicts of interest.

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## Nomenclature

|       |                                        |
|-------|----------------------------------------|
| $A$   | Loaded surface area (mm <sup>2</sup> ) |
| $D$   | Specimen diameter (mm)                 |
| $f_c$ | Compressive strength (MPa)             |
| $f_t$ | Splitting tensile strength (MPa)       |
| $HR$  | Healing ratio (%)                      |
| $L$   | Specimen length (mm)                   |
| $P$   | Maximum applied load (N)               |
| $WA$  | Water absorption (%)                   |
| $W_d$ | Dry specimen weight (g)                |
| $W_h$ | Healed crack width (mm)                |
| $W_i$ | Initial crack width (mm)               |
| $W_s$ | Saturated specimen weight (g)          |

## Greek symbols

|       |                                    |
|-------|------------------------------------|
| $\pi$ | Mathematical constant pi (3.14159) |
|-------|------------------------------------|